------------------------- Slip 13 --------------------

Q.1) Write a C program that illustrates suspending and resuming processes using signals

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        exit(1);
    }

    if (pid == 0) {
        // Child process
        printf("Child process started. PID = %d\n", getpid());
        int i = 0;
        while (1) {
            printf("Child running: i = %d\n", i++);
            sleep(1);
        }
    } else {
        // Parent process
        sleep(3); // Let child run a bit

        printf("Parent suspending child...\n");
        kill(pid, SIGSTOP); // Suspend child
        sleep(5);

        printf("Parent resuming child...\n");
        kill(pid, SIGCONT); // Resume child
```

```
        sleep(5);
        printf("Parent terminating child...\n");
        kill(pid, SIGKILL); // Kill child

        wait(NULL); // Wait for child to terminate
        printf("Child process terminated.\n");
    }

    return 0;
}
```

Q.2) Write a C program that a string as an argument and return all the files that begins with that name in the current directory. For example > ./a.out foo will return all file names that begins with foo

```
#include <stdio.h>
#include <stdlib.h>
#include <dirent.h>
#include <string.h>

int main(int argc, char *argv[]) {
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <prefix>\n", argv[0]);
        return 1;
    }

    char *prefix = argv[1];
    DIR *dir = opendir(".");
    if (!dir) {
        perror("opendir");
        return 1;
    }

    struct dirent *entry;
    printf("Files starting with '%s':\n", prefix);
```

```
    while ((entry = readdir(dir)) != NULL) {
        if (strncmp(entry->d_name, prefix, strlen(prefix)) == 0) {
            printf("%s\n", entry->d_name);
        }
    }

    closedir(dir);
    return 0;
}
```